

□. The Main Technical Parameter of Product.

<i>Properties of PVB film</i>		
Density (g/cm³)	1.069 g/cm³	
Transmittance	>88.0%	(Note 1)
Refractive Index (%)	1.47-1.50	(Note 1)
Tensile Strength (Mpa)	>20	(Note 2)
Elongation at Break (%)	>200	(Note 2)
Humidity (%)	0.35-0.55	(Note 3)
Shrinkage Rate (%)	<6	(Note 4)
Test of Strike	3-9	(Note 5)
Deviation of Width	Expected Value + 10	

Note:

1. Clear float glass (2mm) + Interlayer film (0.76mm) + Clear float glass (2mm)
2. 20□ and humidity 40%
3. Humidity is according as customer requirements
4. 60□, and 40% humidity after 15 minutes to test
5. The HI (Automotive windshields) and ST (Architectural glass) are different interlayer film, the both PVB samples should be tested with -18□ after 2 hours.

□. Packing and Transportation.

1. Products marks and packing:

Marks: Serial numbers of product, number of roll, size, guarantee period, date of production, factory name and contacts.

Packing: PVB film should be separated by PE release film between two layers of PVB film; Vacuum and aluminium packing into wooden case or paper carton. The qualified products have to match guarantee card, feedback card and basic information of merchandise.

2. Storage & transportation:

Storage: Products should be stored in clean warehouse, avoid sunlight to irradiate products directly. It should be stored at temperature of $20^{\circ}\text{C} \pm 5^{\circ}\text{C}$, humidity is 20%-30% if package was opened.

Transportation: No strong sunlight, rain and shake-out while in transportation. The maximum height should no less than 3 floors.

□. The Technical Parameters of PVB Interlayer Film in Producing Process

Step 1: Cutting, Cleaning and Pretreatment.

The cutting trace along the glass border should be proper in order to get a better peeling position. The dimension of glass panel should be precise, with deviation no more than 2 mm, in case the ragged border causing air bubbles. After cutting, the glass border should be edged, and cleaned by glass-washing machine. After cleaning, there should be no contamination like greasy on the glass panels. The endgame of cleaning needs softening water to rinse, in case of being wasted for lacking of adhesive force. The cleaned glass can only be used after it is entirely dried and cooled down to normal temperature.

Requirement of Laminating Room:

Environment of laminating room is very important to manufacture laminated glass. Qualified laminating room should be able to follow the below items.

- The double doors should not open in the meantime and better have air shower equipment in the middle of double doors.
- The plant temperature should keep on 18-20℃ and humidity 20%-30%.
- Wear antistatic whole suit, glove, headgear.
- Clean the equipment and plant in regular time.

Step 2: Laminating Process.

In a qualified lamination room, evenly spread out PVB film on one glass panel and then assemble two glass sheets. Use knife to prune and cut off PVB film edges. During the pruning, it is forbidden to forcibly draw the film, so as to avoid PVB film's distortion. The border of PVB film should be 2-5 mm longer than that of glass sheets. And when pruning, the knife should not touch the glass, so as to avoid air bubbles caused by glass particles around the perimeter of processed laminated glass.

Step 3: Preheating and Exhausting.

The laminated glass needs preheating and exhaust process to press out the leftover air and get a well-edged sealing. Then it can be pressed with high pressure into the required form. The Pre-Nip Process can be divided into two methods:

• Hot Roller Exhausting.

Put the warmed-up laminated glass into a calander. After the first rolling, put it into a thermostat container. After the second rolling, exhaust the air and seal the edges then finish this process. After the second rolling, seal a transparent band around the laminated glass which well-seals the edge in case the air in the high pressure autoclave should stream back and cause air bubbles. Nontransparent areas distributed evenly in the other parts are allowable. This method of using single thermostat container and double rubber roller squeezer requires strictly-controlled temperature on the surface of the glass. It can neither be too high (in case it makes the edges sealing too early to allow the interior air exhaust successfully), nor too low (in case it makes the edge sealing imperfect which causes back-steaming air bubbles).

Suggestion: Keep the surface temperature of the laminated glass within 25-40℃. The space of the first rubber-roller squeezing process should be about 1 mm less than the laminated glass' thickness. The pressure should between 0.2-0.5MPa. The second roller squeezing process should be about 2mm less than the laminated glass' thickness, and the pressure is 0.3-0.5Mpa. For the second roller squeezing, the surface temperature of the glass must be strictly controlled within 60-80℃. When manufacturing, prepare all the glass in advance and try the best to do the process continuously and avoid long waiting time intervals, in case the temperature in the thermo-stated container changes too much so that the surface temperature cannot be controlled precisely.

• **Vacuum Exhausting.**

Put the laminated glass into a vacuum bag or vacuum rubber gasket. With cool pumping and heat pumping, exhaust the air between glass and PVB film layers by the pressure difference and seal the edges.

Suggestion: Decompression vacuum degree should beyond 650 mmHg, the temperature of cool pumping should be under 25℃, and the time should be more than 15 minutes. During heat pumping, the surface temperature of the glass should reach 70-120℃ and be kept for more than 30 minutes.

Step 4: Autoclaving Process.

Put the air-exhausted glass bracket vertically, or with an angle less than 10°. The distance between laminated glass should be more than 5 mm. Close the door after the laminated glass is put into the autoclave, and first raise the temperature to 45℃, then raise the temperature and the pressure at the same time. Control the temperature rising speed at 5℃ per minute, and the pressure adding speed at 0.06 Mpa per minute. When the temperature reaches 120-135℃ and pressure up to 1.0-1.5Mpa, maintain the same for 30 to 60 minutes. After that, keep the same pressure and lower the temperature to 45℃, then lower pressure to atmospheric pressure. This finishes the process of autoclaving and the laminated glass is ready. The temperature must be controlled strictly when rising and lowering pressure in case of air bubbles in the edge of laminated glass.

V. Troubleshooting.

Problem	Reason	Solution
Small bubbles at the edge of laminated safety glass	1) The autoclave in the course of final decrease pressure, high temperature in the autoclave, sometimes the temperature shows under 50℃, but because of different thickness, size of the glass, it will cause temperature of the glass surface to be still very high, so the bubble will be out from the side of the glass, especially the top of the glass that be placed by vertical is easy to appear bubble.	1) Control the pressure time has very good effects to solve this problems, when the temperature shows 50℃ or lower in autoclave, do not hurry in reducing pressure; you should waiting for a while then reduce pressure after being balanced temperature in autoclave, waiting time depends on the equipment, quality, thickness, size and layout of glass.
	2) Smooth degree of tempered glass is bad or PVB interlayer is too thin.	2) Select high quality float glass or use smooth tempered glass, increase the thickness of PVB interlayer.
	3) The temperature and pressure of the autoclave shows inaccurately.	3) Adjust to temperature and pressure in autoclave whether it is accurate or not.
Bubbles were distributed in the middle of laminated safety glass	1) Unsuitable pretreatment, especially roller exhaust, It makes early sealed side before exit	1) Adjust the speed and temperature in laminating, make sure can not seal side before come

	second roller or it is not exhaust before sealed while rubble ring exhaustion. 2) The moisture of PVB interlayer is too high. 3) The glass or tempered glass is not smooth enough; PVB interlayer film is too thin.	second roller our, or adjust to temperature and operation speed of rubble ring, in order to fully exhaust. 2) Check moisture of PVB interlayer, humidity of laminating room, glass surface is clean or not after washing, especially washing continuously. 3) Check quality of float glass, check bend degree of tempered glass, increase the thickness of PVB interlayer film, especially for larger and thick glass.
PVB interlayer film appears stripes and glue off	1) Appear folds in laminated glass. 2) The part of PVB is thin, especially edge and drill place of glass, cause PVB interlayer to shrink and have no PVB interlayer in glass.	1) Take out PVB interlayer evenly, make sure there is no folds. 2) Make sure PVB width is more 2mm than glass width, do not make pvb distortion and do not cut it with over strength.
PVB interlayer film appears glue off at edge and drill place.	1) If appears glue off after exit Autoclave out, because it is not smooth at edge and drill place of tempered glass. 2) Appears glue off at edge and drill place when you use.	1) Adjust roughness of tempered glass, nip clip at edge or drill place of glass before enter autoclave. But don't use too big clips, in order to avoid PVB film being change shape. 2) Control strength well when whole glass is installed, you'd better use the wrench with scale, check the glue which use center of two glasses, it was acid glue, and it should be neutral glue.

The laminated safety glass appears irregular distribution and delamination.	1) Low degree of adhesive between PVB interlayer and glass. 2) Thickness of PVB interlayer film is too thin.	1) a. The water quality of washing glass is bad; suggest use soft water under 20us. b. Improper storage and uncloze
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	3) The smooth and radian are improper.	package of PVB interlayer will cause high moisture, it is easy to appear over 0.6%; if over 1%, the PVB interlayer will whiten after autoclave. c. The wrong laminating and pressure, temperature, pressure or time is too short that can't make good adhesive between PVB interlayer and glass. d. The glass surface have the spots, e.g. detergent, oil sludge. 2) Increase thickness of PVB interlayer film. 3) Choose high-quality float glass, adjust to smooth and radian degree of tempered glass.
Laminated safety glass failed in boiling water.	1) Much air bubbles in laminated glass. 2) The moisture of PVB interlayer film is too high.	1) Seal side too early in stage of pressure; the air bubbles are still staying in laminated glass, although qualified when exit autoclave out, the bubbles will come out at once through the high temperature and boiling water. 2) Adjust storage condition of the PVB interlayer film, guarantee moisture rate of the PVB interlayer among 0.35-0.55%, unqualified product will come if moisture rate over 1%.